



DESIGN AND ANALYSIS OF ALGORITHMS

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DECREASE AND CONQUER

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Combinatorial Objects

- Permutations
- Combinations
- Subsets of a given set

Generating Permutations

- Underlying set elements are to be permuted
- Decrease and conquer approach
- Satisfies the minimal change requirement
- Example: Johnson- Trotter algorithm

Generating Permutations

ALGORITHM *JohnsonTrotter(n)*

//Implements Johnson-Trotter algorithm for generating permutations

//Input: A positive integer n

//Output: A list of all permutations of $\{1, \dots, n\}$

initialize the first permutation with $\overleftarrow{1} \overleftarrow{2} \dots \overleftarrow{n}$

while the last permutation has a mobile element **do**

 find its largest mobile element k

 swap k with the adjacent element k 's arrow points to

 reverse the direction of all the elements that are larger than k

 add the new permutation to the list

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Decrease and Conquer

←←←← 1 2 3 4 →←←←	←←←← 1 2 4 3 ←→←←	←←←← 1 4 2 3 ←←→←	←←←← 4 1 2 3 ←←←→
4 1 3 2 ←←←←	1 4 3 2 ←←←←	1 3 4 2 ←←←←	1 3 2 4 ←←←←
3 1 2 4 →→←←	3 1 4 2 →→←←	3 4 1 2 →←→←	4 3 1 2 →←←→
4 3 2 1 ←→←←	3 4 2 1 ←→←←	3 2 4 1 ←←→←	3 2 1 4 ←←→←
2 3 1 4 →←←→	2 3 4 1 ←→←→	2 4 3 1 ←←→→	4 2 3 1 ←←→→
4 2 1 3	2 4 1 3	2 1 4 3	2 1 3 4

Generating Subsets:

Knapsack problem needed to find the most valuable subset of items that fits a knapsack of a given capacity.

Powerset: set of all subsets of a set. Set $A = \{1, 2, \dots, n\}$ has 2^n subsets.

Generate all subsets of the set $A = \{1, 2, \dots, n\}$.

Any **decrease-by-one** idea?

of subsets of $\{ \}$ = $2^0 = 1$, which is $\{ \}$ itself

Suppose, we know how to generate all subsets of $\{1, 2, \dots, n-1\}$

Now, how can we generate all subsets of $\{1, 2, \dots, n\}$?

Generating Subsets:

All subsets of $\{1, 2, \dots, n-1\}$: 2^{n-1} such subsets

All subsets of $\{1, 2, \dots, n\}$:
 2^{n-1} subsets of $\{1, 2, \dots, n-1\}$ and
another 2^{n-1} subsets of $\{1, 2, \dots, n-1\}$ having '**n**' with them.

That adds up to all 2^n subsets of $\{1, 2, \dots, n\}$

0	$\emptyset$							
1	$\emptyset$	$\{a_1\}$						
2	$\emptyset$	$\{a_1\}$	$\{a_2\}$	$\{a_1, a_2\}$				
3	$\emptyset$	$\{a_1\}$	$\{a_2\}$	$\{a_1, a_2\}$	$\{a_3\}$	$\{a_1, a_3\}$	$\{a_2, a_3\}$	$\{a_1, a_2, a_3\}$

Alternate way of Generating Subsets:

Knowing the binary nature of either having n th element or not, any idea involving binary numbers itself?

One-to-one correspondence between all 2^n bit strings $b_1b_2\dots b_n$ and 2^n subsets of $\{a_1, a_2, \dots, a_n\}$.

Each bit string $b_1b_2\dots b_n$ could correspond to a subset.

In a bit string $b_1b_2\dots b_n$, depending on whether b_i is 1 or 0, a_i is in the subset or not in the subset.

000	001	010	011	100	101	110	111
$\emptyset$	$\{a_3\}$	$\{a_2\}$	$\{a_2, a_3\}$	$\{a_1\}$	$\{a_1, a_3\}$	$\{a_1, a_2\}$	$\{a_1, a_2, a_3\}$

Generating Subsets in Squashed order:

Squashed order: any subset involving a_j can be listed only after all the subsets involving $a_1, a_2, \dots, a_{j-1}$

Both of the previous methods does generate subsets in squashed order.

000	001	010	011	100	101	110	111
$\emptyset$	$\{a_3\}$	$\{a_2\}$	$\{a_2, a_3\}$	$\{a_1\}$	$\{a_1, a_3\}$	$\{a_1, a_2\}$	$\{a_1, a_2, a_3\}$

Generating Subsets in Squashed order:

Squashed order: any subset involving a_j can be listed only after all the subsets involving $a_1, a_2, \dots, a_{j-1}$

Can we do it with minimal change in bit-string (actually, just one-bit change to get the next bit string)? This would mean, to get a new subset, just change one item (remove one item or add one item).

Binary reflected gray code:

000 001 011 010 110 111 101 100



THANK YOU

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